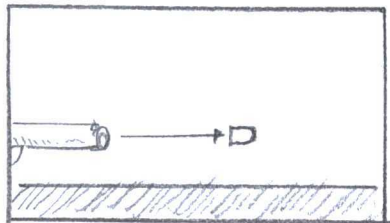


Chapter 3: Momentum and Angular Momentum:

3.1



"gun... fires shell...
above ground"

(pg 83) "Principle of Conservation of Momentum"

$$\vec{P} = \sum m_{\alpha} \vec{V}_{\alpha} \quad \text{where } P = \text{"Total Momentum"}$$

Total momentum = ~~sum~~ (m = "mass")

v = "velocity"

$$(M+m) v_1 = M \vec{v}_1 + m v_2$$

$$v = \frac{m}{M+m} v_2$$

$$= \frac{1}{\frac{M}{m} + 1} v_2$$

△ An incorrect
typeset. from
unclear velocities.

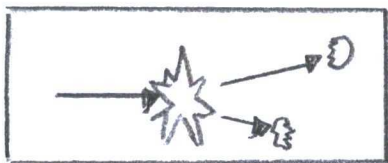
v = velocity relative
to ground

M = Mass of gun

m = Mass of bullet

v_2 = muzzle speed.

3.2.



"A shell traveling...
explodes into two...
fragments"

(pg 83) "Principle of Conservation of Momentum"

$$P = \sum m_{\alpha} v_{\alpha}$$

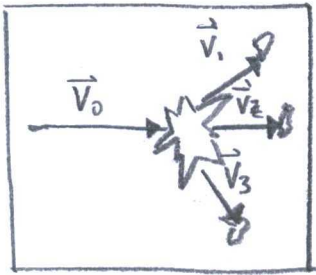
$$\underbrace{(m_1 + m_2) v_0}_{\text{"Initially"}} = \underbrace{m_1 v_0 + m_2 v_0}_{\text{"After"}}$$

$$2 m v_0 = m_1 v_0 + m_2 v_0 \dots \text{when } m_1 = m_2 = m$$

$$m_1 = m_2 = m$$

$$v_0 = v_0$$

3.3.



"shell... explodes into three pieces"

(pg 83) "Principle of Conservation of Momentum"

$$P = \sum m_{\alpha} v_{\alpha}$$

$$(m_1 + m_2 + m_3) v_0 = m_1 v_1 + m_2 v_2 + m_3 v_3$$

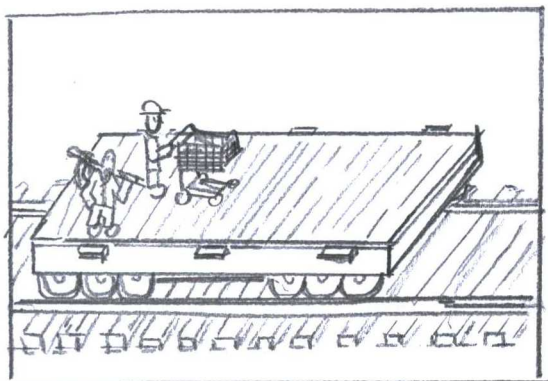
If $m_1 = m_2 = m_3$ and $v_1 = v_0$ and $v_2 = v_3$,

then $3 m v_0 = m v_0 + 2 m v_1$

$$v_0 = v_1$$

3.4

a) (pg 83) "Principle of Conservation of Momentum"



"Two hobos... Standing at one end of a stationary railroad flatcar"

$$P = \sum m_{\alpha} v_{\alpha}$$

$$\underbrace{(m_h + m_h + m_{fc}) V_0}_{\text{"Initial"}} = \underbrace{m_h u + m_h u + m_{fc} V_f}_{\text{"Final"}}$$

when m_h = mass hobo

m_{fc} = mass flatcar

V_0 = stationary flatcar

u = hobo speed off flatcar

V_f = flatcar final velocity.

$$\text{If } V_0 = 0, \text{ then } V_f = \frac{-2m_h u}{m_{fc}}$$

b) If $V_0 = 0$, then initially momentum is zero, ~~but~~ not at final

$$0 = m_h u + m_h v + m_{fc} V$$

But final velocity changes after consecutive jumping.

$$V = \frac{-m_h}{m_h + m_{fc}} u$$

Then second hobo jumps,

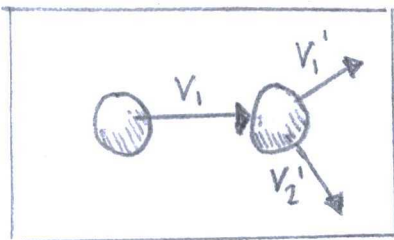
$$(m_h + m_{fc}) \cdot V = m_h \cdot u + m_{fc} \cdot V_f$$

$$- \frac{(m_h + m_{fc}) m_h u}{(m_h + m_{fc})} = m_h \cdot u + m_{fc} \cdot V_f$$

$$V_f = - \frac{2 m_h \cdot u}{m_{fc}}$$

Whether both hobos jumped at the same instant or separate instances, velocity remained constant.

3.5



"Two billiard balls"

Vector Equation:

$$m_1 \vec{v}_1 + m_2 \vec{v}_2^0 = m_1 \vec{v}_1' + m_2 \vec{v}_2'$$

$$m_1 = \frac{m_1 \vec{v}_1' + m_2 \vec{v}_2'}{\vec{v}_1}$$

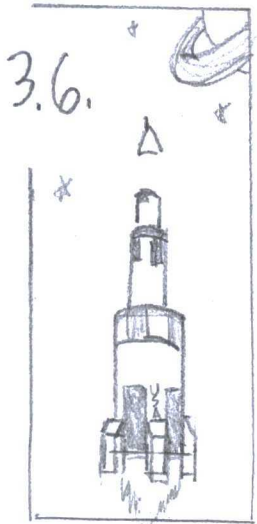
Scalar Equation:

$$m_1 = \frac{m_1 (\vec{v}_1 \times \vec{v}_2')}{\vec{v}_1}$$

$$= \frac{m_1 (|\vec{v}_1| |\vec{v}_2'| \sin \theta)}{\vec{v}_1}$$

$$= \frac{m_1 |\vec{v}_1| |\vec{v}_2|}{|\vec{v}_1|}$$

only by the equal masses



"Saturn V

Rocket launch"

Free-body Diagram



Total momentum:

(pg 86) "Total momentum (rocket)"-modified"

$$P(t+dt) = mv + d(m \cdot v)$$

$$= mv + m dv + dm v$$

$$m dv = -dm v_{ex}$$

$$m \frac{dv}{dt} = - \frac{dm}{dt} v_{ex}$$

$$m \dot{v} = - \dot{m} v_{ex} \rightarrow$$

$$= \text{Thrust}$$

(from problem) "Parameters"

$\dot{m}$ = "mass ejected"

$$= 15,000 \text{ Kg/s}$$

$$V_{ex} = 2,500 \text{ m/s}$$

$$\text{Thrust} = -\dot{m} V_{ex}$$

$$= -15,000 \text{ kg/s} \cdot 2,500 \text{ m/s}$$

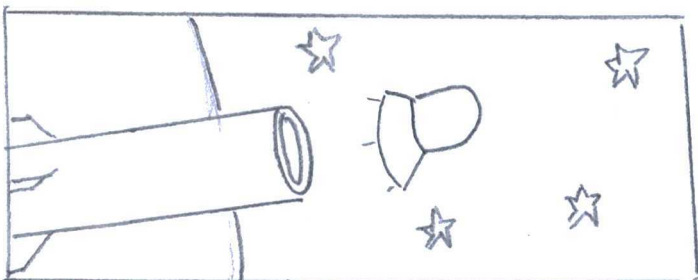
$$= -3.75 \times 10^7 \text{ Newtons}$$

A comparison to 3000 tons:

$$3000 \text{ tons} \times \frac{1000 \text{ Kg}}{1 \text{ ton}} \times \frac{9.8 \text{ m}}{\text{s}^2} = 2.94 \times 10^7 \text{ N}$$

$$\frac{|-3.75 \times 10^7 \text{ N}|}{2.94 \times 10^7 \text{ N}} = 1.28 \text{ multiples force.}$$

3.17



"launch of a space shuttle ... in outer space"

Shuttle speed:

(3.7) "Thrust" - modified

$$m \dot{v} = -\dot{m} v_{ex}$$

$$\int_{v_0}^v dv = -v_{ex} \int_{m_0}^m \frac{1}{m} dm$$

$$v = v_{ex} \ln(m_0/m) + v_0$$

$$= 3,000 \frac{m}{s} \ln\left(\frac{2 \times 10^6 \text{ Kg}}{1 \times 10^6 \text{ Kg}}\right) + 0$$

$$\approx +2,000 \frac{m}{s}$$

Thrust:

(3.7) "Thrust"

$$\text{Thrust} = -\dot{m} v_{ex}$$

$$= -\frac{(m_f - m_0)}{(t_f - t_0)} v_{ex}$$

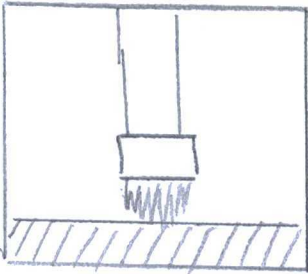
$$= -\frac{(2 \times 10^6 \text{ Kg} - 1 \times 10^6 \text{ Kg})}{(2 \text{ min} - 0)} \cdot 3000 \frac{m}{s}$$

$$= 2.5 \times 10^7 \text{ N}$$

Comparison to Initial Weight:

$$\frac{2.5 \times 10^7 \text{ N}}{2.0 \times 10^7 \text{ N}} = 1.25 \text{ multiples initial force.}$$

3.8.



"Rocket ...
hovers stationary"

a) (3.7) "Thrust"-modified"

$$m \dot{v} = -\dot{m} v_{ex}$$

$$m \dot{v} \int_0^t dt = -v_{ex} \int_{m_0}^m \frac{dm}{m} \quad \text{... when } \dot{v} = \text{constant}$$

$$t = -\frac{v_{ex}}{a_g} \ln(m/m_0)$$

When v_{ex} = "velocity thrusters"
 a_g = "acceleration by gravity"

m = "current mass"
 $= \lambda m_0$

m_0 = "initial mass"

$$t = -\frac{v_{ex}}{a_g} \ln(\lambda m_0/m_0)$$

$$= -\frac{v_{ex}}{a_g} \ln(\lambda)$$

b) If $v_{ex} \approx 3000 \text{ m/s}$ and $\lambda \approx 10\%$

$$\text{then } t = \frac{-3000 \text{ m/s}}{9.8 \text{ m/s}^2} \ln(0.10)$$

$$= 7 \times 10^2 \text{ seconds}$$

3.9

(3.6) "Thrust"

$$m \dot{v} = -\dot{m} v_{ex} \quad \text{where } m = \text{Initial mass} = 2 \times 10^6 \text{ kg}$$

$$v_{ex} = -\frac{m \dot{v}}{\dot{m}}$$

$$= -\frac{2 \times 10^6 \text{ kg} \cdot 9.8 \text{ m/s}^2}{-8.3 \times 10^3 \text{ kg/s}}$$

$$\approx 2.4 \times 10^3 \text{ m/s}$$

$\dot{v}$ = Acceleration by gravity

$$= 9.8 \text{ m/s}^2$$

$-\dot{m}$ = Ejecting mass

$$= \frac{\Delta m}{\Delta t}$$

$$\approx -8.3 \times 10^3 \text{ kg/s}$$

3.10

Maximum momentum in outer space:

(pg 86) "Total momentum"

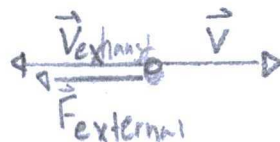
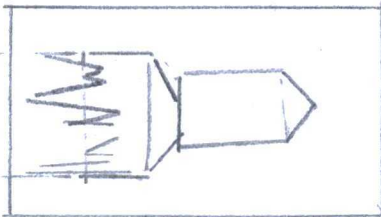
$$\begin{aligned} P(t+dt) &= (m+dm)(v+dv) - dm(v-v_{ex}) \\ &= mv + m dv + dm \overset{\circ}{v} + dm \overset{\circ}{v} dt - dm \overset{\circ}{v} + dm v_{ex} \end{aligned}$$

$$\begin{aligned}
 &= m v + m dv + dm v_{ex} \\
 \frac{dP}{dt} &\approx \frac{P(t+dt) - P(t)}{dt} \\
 &= \frac{m dv + dm v_{ex}}{dt} \\
 &= m \dot{v} + \dot{m} v_{ex} \\
 &= 0
 \end{aligned}$$

Maximum momentum occurs
at $m \dot{v} = -\dot{m} v_{ex}$, so initially.

3.11

Free-body Diagram:



"Rocket... subject
to external force"

a) Equation of Motion:

$$\begin{aligned}
 \vec{F} &= m \frac{d\vec{v}}{dt} = -\frac{d}{dt}(m \cdot \vec{v}) + \vec{F}_{external} \\
 &= -\dot{m} \vec{v} - m \frac{d\vec{v}}{dt} + \vec{F}_{external} \\
 &= -\dot{m} \vec{v} + \vec{F}_{external}
 \end{aligned}$$

b) If $\vec{F}_{ext} = \vec{F}_g$, then $m \dot{\vec{v}} = -\dot{m} \vec{v} - m \vec{g}$
Velocity: as a function of t:

$$\begin{aligned}
 m \frac{dv}{dt} &= -\dot{m} v_{ex} + mg \\
 \int_{v_0}^v dv &= \int_{t_0}^t -\frac{\dot{m}}{m} v_{ex} dt - \int_{t_0}^t g dt \\
 &= \int_{t_0}^t \left(-\frac{(m-m_0)}{t} \right) v_{ex} dt - \int_{t_0}^t g dt \\
 &= \int_{t_0}^t -\left(1 - \frac{m_0}{m}\right) \frac{1}{t} v_{ex} dt - (t-t_0)g
 \end{aligned}$$

$$V - V_0 = -v_{ex} \left(1 - \frac{m_0}{m}\right) \ln(t/t_0) - (t-t_0)g$$

$$V = -v_{ex} \left(1 - \frac{m_0}{m}\right) \ln(t/t_0) + (t_0 - t)g + V_0$$

c) Speed two minutes into flight:

(From problem 3.7) m_0 = "Initial mass"

$$= 2 \times 10^6 \text{ Kg}$$

m = "Final mass"

$$= 1 \times 10^6 \text{ Kg}$$

v_{ex} = "Exhaust speed"

$$= 3000 \text{ m/s}$$

V_0 = "Initial velocity"

$$= 0$$

$$V = -3000 \frac{\text{m}}{\text{s}} \left(1 - \frac{2 \times 10^6 \text{ kg}}{1 \times 10^6 \text{ kg}} \right) \ln(120) - 120 \times 9.8 \frac{\text{m}}{\text{s}^2} + 0$$

$$\approx 103000 \frac{\text{m}}{\text{s}}$$

If no gravity, still around 10,000 m/s.

d) If $-\dot{m}v_{\text{ex}} < -mg$, then no movement occurs by the rocket.

3.12.

a) 1st stage during lift:

(3.6) "Thrust"

$$m\dot{v} = -\dot{m}v_{\text{ex}}$$

$$v = -v_{\text{ex}} \ln(m/m_0) + v_0$$

$$= -v_{\text{ex}} \ln(0.6) + \dots \text{When } m = 0.60m_0$$

$$v_0 = 0$$

$$= +0.5 \times v_{\text{ex}}$$

b) 2nd stage during lift:

$$v_1 = -v_{\text{ex}} \ln(m_1/m_0) + v_0$$

$$= +0.4 \cdot v_{\text{ex}}$$

... from $m_1 = 0.7m_0$

$$v_2 = -v_{\text{ex}} \ln(0.3) + 0.4v_{\text{ex}} \dots \text{by } m_2 = 0.4m_0$$

$$\approx 2 \cdot v_{\text{ex}}$$

$$@ t=0, v=0, \text{ so } C = \frac{R V_{ex}}{b}$$

$$v = \frac{R V_{ex}}{b} (1 - e^{-bt/m})$$

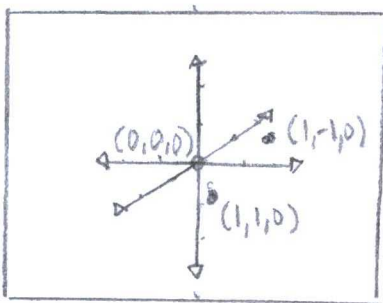
$$= \frac{R V_{ex}}{b} (1 - e^{\frac{-b(m_0 - m)}{R m}})$$

$$= \frac{R V_{ex}}{b} (1 - e^{(1 - m/m_0)^{b/R}})$$

$$= \frac{R V_{ex}}{b} (1 - (1 - 1 + m/m_0 + \dots)^{b/R})$$

$$= \frac{R V_{ex}}{b} (1 - (m/m_0)^{b/R})$$

3.15



"center of mass"

Center of mass:

(3.9) "Center of mass"

$$R = \frac{1}{M} \sum_{\alpha=1}^N m_{\alpha} r_{\alpha} = \frac{m_1 r_1 + \dots + m_N r_N}{M}$$

$$= \frac{1}{m_1 + m_2 + m_3} \sum_{\alpha=1}^3 m_{\alpha} r_{\alpha}$$

$$= \frac{m_1 \cdot 1 + m_2 \cdot 1 + m_3 \cdot 0}{m_1 + m_2 + m_3}$$

$$= 2m_1 / 12m_1 = 1/6$$

Velocity increased by four-fold
from multiple stages.

3.13

a) (from Exercise 3.11) "velocity"

$$V = V_{ex} \ln\left(\frac{m_0}{m_0 - kt}\right) - g t + V_0$$

$$\int_0^y dy = V_{ex} \int_0^t \ln\left(\frac{m_0}{m_0 - kt}\right) dt - \int_0^t g t dt + V_0 \int_0^t dt$$

$$= -\frac{m V_{ex}}{k} \ln\left(\frac{m_0}{m}\right) - \frac{1}{2} g t^2 + V_{ex} t$$

3.14

(from exercise) "Linear Resistive Force"

$$\vec{F} = -b\vec{V}$$

(3.29) "Equation of motion"

$$m\dot{v} = -mV_{ex} + F^{ext}$$

$$= -mV_{ex} - b\vec{V}$$

$$\int_0^V \frac{dv}{kV_{ex} + bv} = \int_0^t \frac{1}{m} dt$$

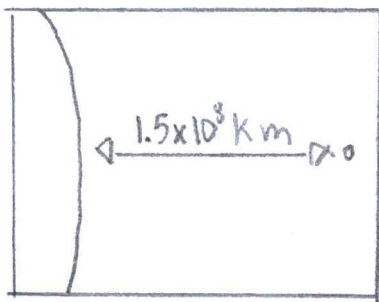
$$-\ln(kV_{ex} + bv) = \frac{t}{m} + C$$

Other centers, $R_y = 0$ & $R_z = 0$

$$\vec{R} = \langle 1/6, 0, 0 \rangle$$

The point near the origin has
"planetary" mass.

3.16.



"Earth and sun"

Center of mass:

(3.9) "center of mass"

$$R = \frac{1}{M} \sum_{\alpha=1}^N m_{\alpha} r_{\alpha} = \frac{m_1 r_1 + \dots + m_N r_N}{M}$$

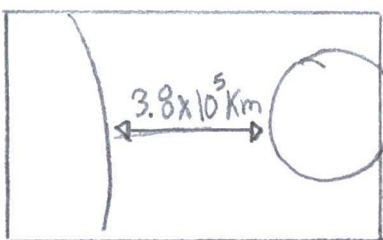
$$= \frac{M_e \cdot r_e + M_s \cdot r_s}{M_e + M_s}$$

$$\approx \frac{6.0 \times 10^{24} \text{ kg} \cdot 1.5 \times 10^8 \text{ km} + 2.0 \times 10^{30} \text{ kg} \cdot 0}{6.0 \times 10^{24} \text{ kg} + 2.0 \times 10^{30} \text{ kg}}$$

$$\approx 4.5 \times 10^2 \text{ km}$$

↑ ...less than the sun's radius
at $7.0 \times 10^5 \text{ km}$.

3.17. "Earth and moon"



center of mass:

(3.9) "center of mass"

$$R = \frac{1}{M} \sum_{\alpha=1}^N m_{\alpha} r_{\alpha} = \frac{m_1 r_1 + \dots + m_N r_N}{M}$$

$$= \frac{M_e r_e + M_m r_m}{M_e + M_m}$$

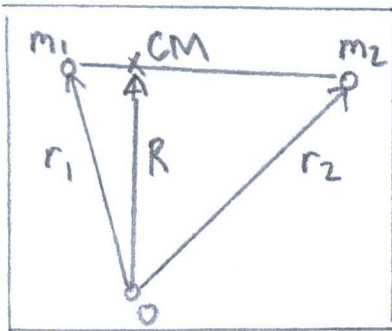
$$\cong \frac{6.0 \times 10^{24} \text{ Kg} \cdot 0 + 7.4 \times 10^{22} \text{ Kg} \cdot 3.8 \times 10^5 \text{ Km}}{6.0 \times 10^{24} \text{ Kg} + 7.4 \times 10^{22} \text{ Kg}}$$

$$\cong 4.6 \times 10^3 \text{ Km}$$



much less than the earth's radius at $6.4 \times 10^3 \text{ Km}$.

3.10



"center of mass"

a) Vector #1: $\vec{r}_m = \vec{R} - \vec{r}_1 = \frac{(m_1 - M)\vec{r}_1 + m_2 \vec{r}_2}{M}$

Vector #2: $\vec{r}_m = \vec{R} - \vec{r}_2 = \frac{m_1 \vec{r}_1 + (m_2 - M)\vec{r}_2}{M}$

b) Ratios m_2/m_1 :

Vector #1: $\vec{r}_m = \left(\frac{m_1}{m_1 + m_2} - 1\right)\vec{r}_1 + \left(\frac{m_2}{m_1 + m_2}\right)\vec{r}_2$
 $= \left(\frac{m_2}{m_1 + m_2}\right)[\vec{r}_2 - \vec{r}_1]$

Vector #2: $\vec{r}_m = \frac{m_1 \vec{r}_1 + (m_2 - M)\vec{r}_2}{M}$

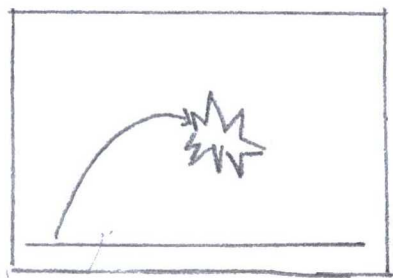
$$= \left(\frac{m_1}{m_1 + m_2}\right)[\vec{r}_1 - \vec{r}_2]$$

If $m_1 \gg m_2$:

Vector #1: $\vec{r}_m \cong \left(\frac{m_2}{m_1}\right)[\vec{r}_2 - \vec{r}_1]$ ← "marginally close to m_1 "

Vector #2: $\vec{r}_m \cong \vec{r}_1 - \vec{r}_2$

3.19.



"projectile exploded in midair."

a) (3.12) "Total Force"

$$F^{\text{ext}} = M\ddot{R}$$

During the explosion, center of mass remains on the parabola.

An equation for center of mass is $y(t) = y_0 + v_{0y}t - \frac{1}{2}a_y t^2$ and

$x(t) = x_0 + v_{0x}t - \frac{1}{2}a_x t^2$ to position.

b) Projectiles break into pieces during trajectory. If two pieces, where one piece lands 100m beyond target, the other lands 100m before target.

c) Center of mass remains the same irregardless of time.

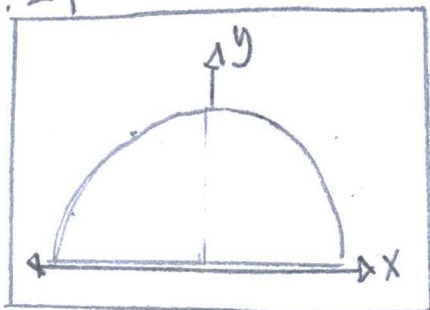
3.20.

For an elastic collision, $r_i \sum_{i=1}^n m_i = \sum_{i=1}^n m_i r_i$

$$\text{So, } r(m_1 + m_2) = m_1 r_1 + m_2 r_2$$

$$r = \frac{m_1 r_1 + m_2 r_2}{(m_1 + m_2)}$$

3.21



"thin sheet of metal cut in shape of a semicircle"

(3.9) "Center of mass"

$$R = \frac{1}{M} \sum_{\alpha=1}^N m_{\alpha} r_{\alpha} = \frac{m_1 r_1 + \dots + m_N r_N}{M}$$

$$= \frac{1}{M} \int \sigma r dA$$

σ = surface mass density (kg/m^2)

$$= \int \frac{r}{A} dA$$

dA = element area ($rd\phi$)

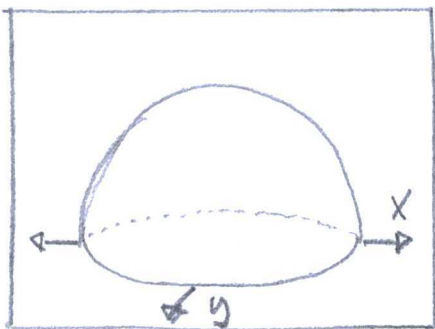
$$= \int_0^R \int_0^{\pi} \frac{r}{\pi R^2} r \sin \phi dr d\phi$$

$$= \int_0^R \frac{r^2}{\pi R^2} dr \int_0^{\pi} \sin \phi d\phi$$

$$= \frac{4}{3\pi} \cdot R \text{ is center of mass.}$$

in polar coordinates.

3.22.

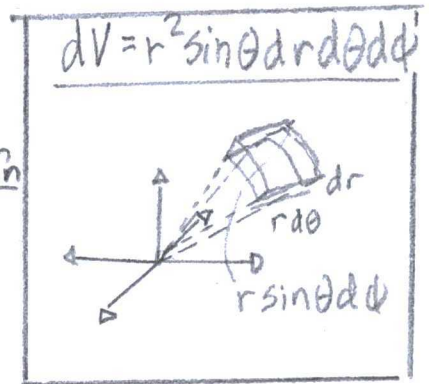


"A uniform solid hemisphere lies in the xy plane with its center at the origin"

Center of mass in spherical coord.

(3.9) "Center of mass"

$$R = \frac{1}{M} \sum_{k=1}^N m_k r_k = \frac{m_1 r_1 + \dots + m_N r_N}{M}$$



$$= \int \frac{z}{V} dV$$

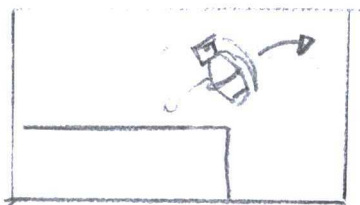
$$= \int \frac{r \cos \phi}{\frac{4}{3} \pi R^3} dV$$

$$= \int_0^R \int_0^\pi \int_0^{2\pi} \frac{r \cos \phi}{\frac{4}{3} \pi R^3} r^2 \sin \theta dr d\theta d\phi$$

$$= \int_0^R \frac{r}{\frac{4}{3} \pi R^3} dr \int_0^\pi \sin \theta d\theta \int_0^{2\pi} d\phi$$

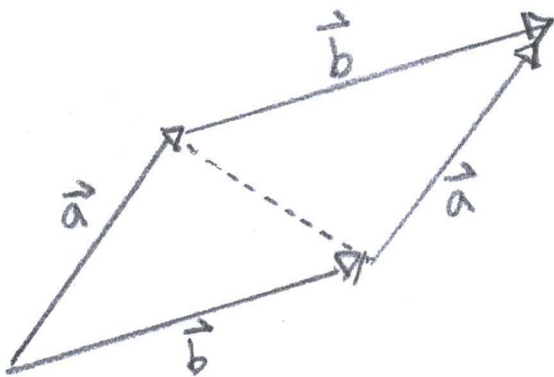
$$= \frac{3}{8} R \text{ is the center of mass.}$$

3.23



"grenade thrown off...
a high cliff"

3.24

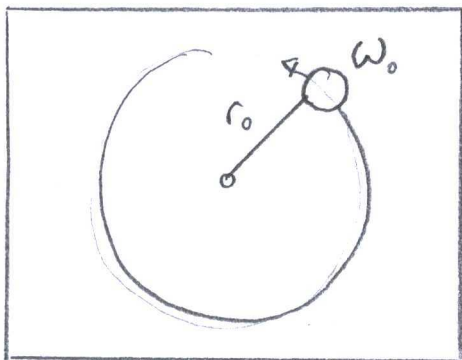


$$\text{Parallelogram Area} = |\vec{a} \times \vec{b}|$$

$$= 2 \cdot (\text{Triangle Area})$$

$$\text{Triangle Area} = \frac{1}{2} |\vec{a} \times \vec{b}|$$

3.25



"particle... moving
in a circle"

Angular Momentum for a single particle:

(3.14) "Angular momentum"

$$l = r \times p$$

where r = radius

p = momentum

$$= r \times mv$$

$$= r \times m(\omega r)$$

$$= m r_0^2 \omega_0^2 \quad \dots \text{initially}$$

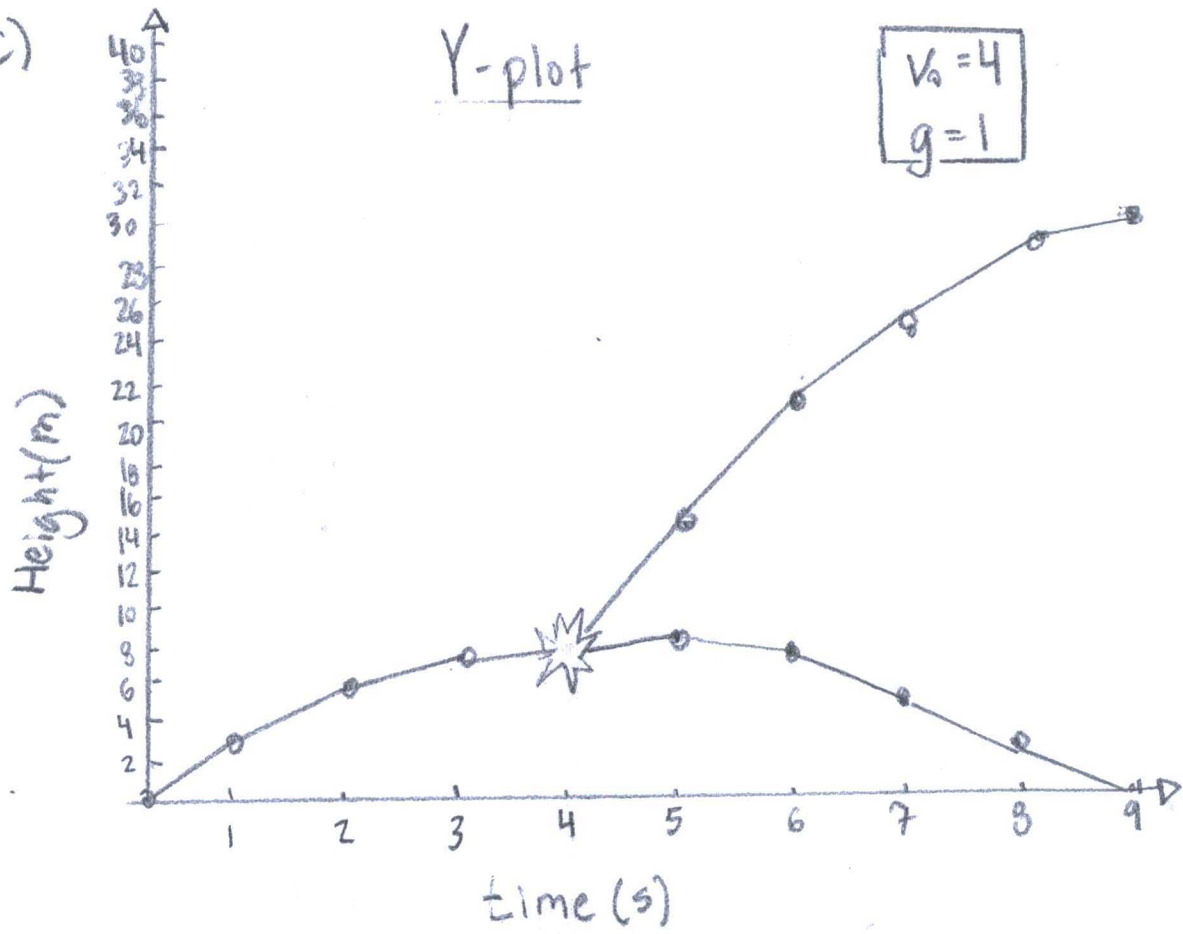
$$= m r^2 \omega^2 \quad \dots \text{now}$$

$$\omega = \frac{r_0}{r} \omega_0 \quad \dots \text{angular velocity now}$$

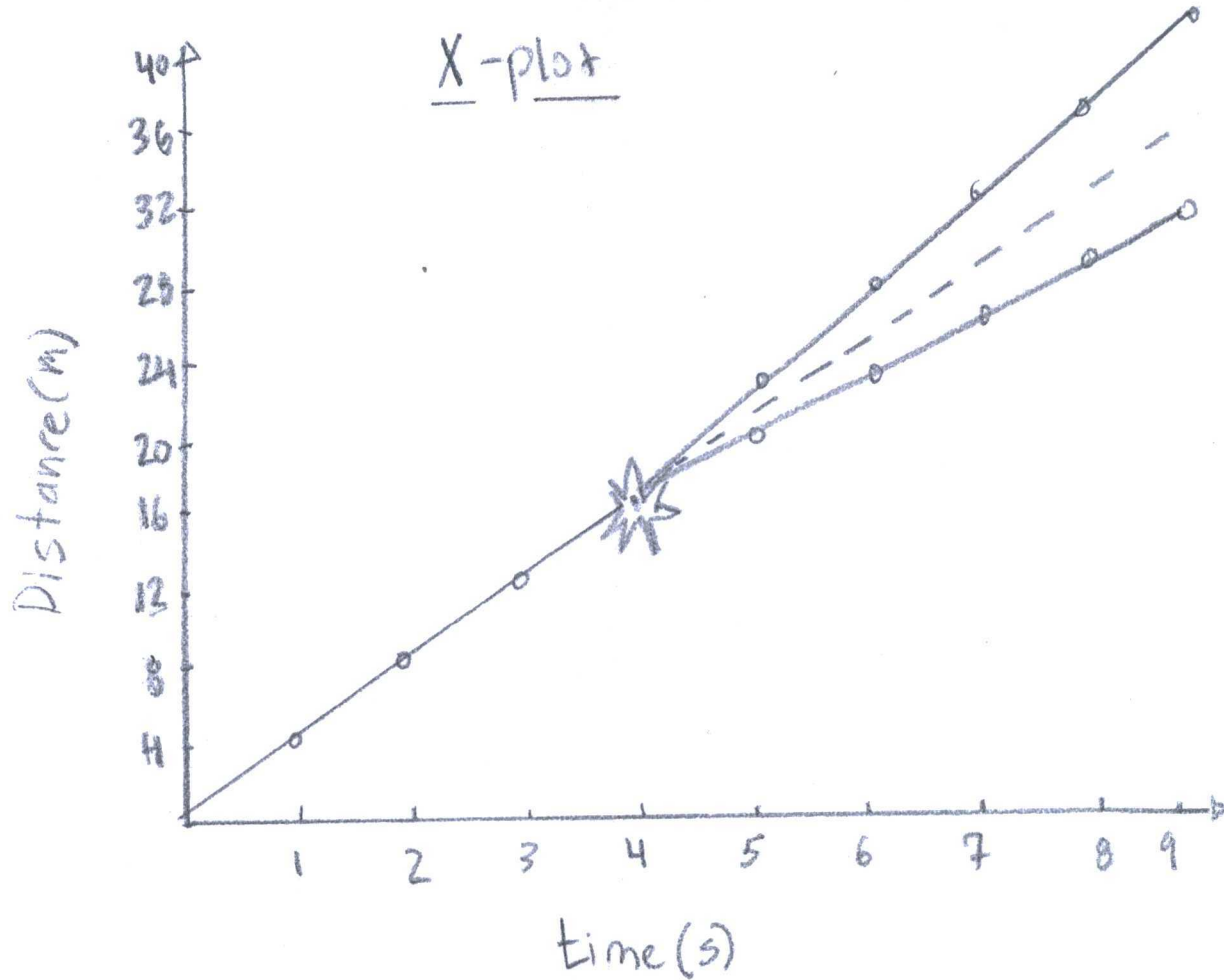
a, b, c)

Y-plot

$V_0 = 4$
 $g = 1$



X-plot



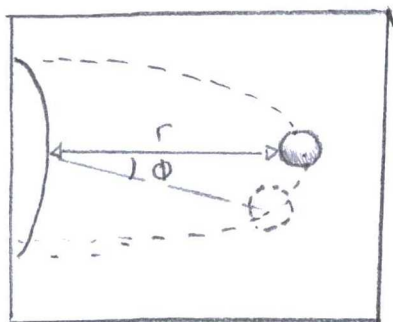
3.26

a) Particles move around a fixed origin O because constant angular momentum. Equation 3.16 $\dot{\mathbf{l}} = \mathbf{r} \times \mathbf{F} = \mathbf{T}$ equals zero or $\dot{\mathbf{l}} = 0$.

With ~~no~~ torque, the particles tend planar motion.

b) Particle's orbit lie in a single plane about the origin. Forces during rotation, centripetal, centrifugal, and torque exist. But torque is zero.

3.27



"Planet orbiting the fixed sun"

a) Angular Momentum:

(3.14) "Angular momentum"

$$\mathbf{l} = \mathbf{r} \times \mathbf{p}$$

$$= \mathbf{r} \times (m\mathbf{v})$$

$$= \mathbf{r} \times (m \cdot \mathbf{r} \omega)$$

$$= m r^2 \omega$$

where $\omega = \dot{\phi}$

b) (pg 92) "Keplers Second Law"

"As each planet moves around the sun, a line drawn from the planet to the sun sweeps out equal areas in equal times"

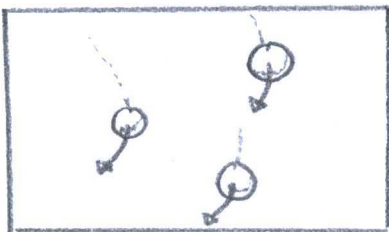
$$\frac{dA}{dt} = \frac{r d\phi \cdot dr}{dt} = \int r(\omega) dr$$

$$= \frac{r^2}{2} \omega$$

$$= \frac{l/m}{2} \quad \dots \text{When } l = mr^2\omega$$

$$= \frac{l}{2m}$$

3.28



"Angular momentum for several particles"

(3.20) "Total Angular momentum"

$$L = \sum_{\alpha=1}^N l_{\alpha} = \sum_{\alpha=1}^3 r_{\alpha} \times p_{\alpha}$$

$$\dot{L} = \frac{d}{dt} L$$

$$= \sum_{\alpha=1}^3 \dot{l}_{\alpha} = \sum_{\alpha=1}^3 r_{\alpha} \times F_{\alpha}$$

$$= \sum_{\alpha=1}^3 r_{\alpha} \times \left(\sum_{\beta \neq \alpha}^3 F_{\alpha\beta} + F_{\alpha}^{\text{ext}} \right)$$

$$= \sum_{\alpha=1}^3 \sum_{\beta \neq \alpha}^3 r_{\alpha} \times F_{\alpha\beta} + \sum_{\alpha=1}^3 r_{\alpha} \times F_{\alpha}^{\text{ext}}$$

$$= \sum_{\alpha=1}^3 \sum_{\beta > \alpha}^3 (r_{\alpha} \times F_{\alpha\beta} + r_{\beta} \times F_{\beta\alpha}) + \sum_{\alpha=1}^3 r_{\alpha} \times F_{\alpha}^{\text{ext}}$$

$$= \sum_{\alpha=1}^3 \sum_{\beta > \alpha}^3 (r_{\alpha} - r_{\beta}) \times F_{\alpha\beta} + \sum_{\alpha=1}^3 r_{\alpha} \times F_{\alpha}^{\text{ext}}$$

$$= \tau^{\text{ext}}$$

3.29



"spherical asteroid
... spinning"

Angular Velocity:

(from Problem) "Moment of Inertia"

$$I = \frac{2}{5} MR^2$$

(3.14) "Angular Momentum"

$$l = r \times p$$

$$= I \omega$$

$$= \frac{2}{5} MR^2 \cdot \omega$$

$$l_0 = \frac{2}{5} M R_0^2 \omega_0$$

$$\frac{L}{L_0} = \frac{\frac{2}{5} M R^2 \omega}{\frac{2}{5} M_0 R_0^2 \omega_0}$$

$$\omega = \frac{M_0}{M} \frac{R_0^2}{R^2} \omega_0$$

Final angular velocity after the radius doubles:

$$\begin{aligned} \omega &= \frac{M_0}{M} \frac{R_0^2}{R^2} \omega_0 \\ &= \frac{M_0}{M} \frac{R_0^2}{(2R_0)^2} \omega_0 \\ &= \frac{M_0}{4M} \omega_0 \end{aligned}$$

3.30

a) Velocity:

(pg = 34) "Cylindrical Polar"

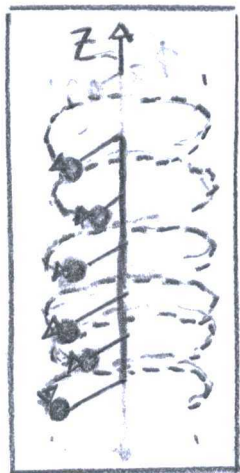
(ρ, ϕ, z)

(1.44) "Rotational velocity"

$$v = \omega \times r = r \omega$$

$$\approx \rho \omega$$

$$\approx \rho \frac{d\phi}{dt} \hat{\phi}$$



"rigid body rotating"

b) Z-component of angular momentum:

(3.14) "Angular momentum"

$$l = r \times p$$

$$\cong r \times p$$

$$\cong \rho \hat{r} \times m v$$

$$\cong \rho \hat{r} \times m \rho \omega \hat{\phi} \quad \dots \text{from part a}$$

$$\cong m \rho^2 \omega (\hat{r} \times \hat{\phi}) \quad \dots \text{cross product between unit vectors.}$$

$$\cong m \rho^2 \omega \hat{z}$$

c) Total Angular Momentum:

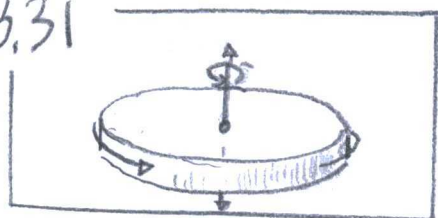
(3.20) "the total angular momentum"

$$L = \sum_{\alpha=1}^N l_{\alpha}$$

$$= \sum_{\alpha=1}^N m_{\alpha} \rho_{\alpha}^2 \omega \hat{z}$$

$$= I \omega \hat{z}$$

3.31



"Uniform disc
rotating about
its axis"

(3.31) "Moment of Inertia"

$$I = \sum_{\alpha=1}^N m_{\alpha} r_{\alpha}^2$$

$$= \int_0^R \int_0^{2\pi} \sigma r^2 r dr d\phi$$

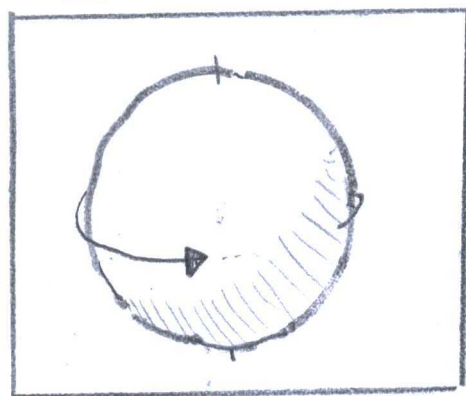
$$= \frac{M}{\pi R^2} \int_0^R r^3 dr \int_0^{2\pi} d\phi$$

$$= \frac{MR^2}{2}$$

σ = "surface
mass
density"

$$= \frac{M}{\pi R^2}$$

3.32.



"Uniform sphere
rotating"

(3.31) "Moment of Inertia"

$$I = \sum_{\alpha=1}^N m_{\alpha} r_{\alpha}^2$$

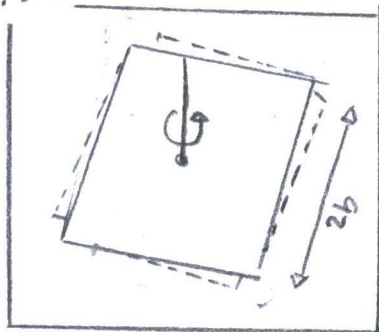
$$= \int_0^R \int_0^{\pi} \int_0^{2\pi} \sigma (r \sin \theta)^2 r^2 \sin \theta dr d\theta d\phi$$

$$= \frac{M}{\frac{4}{3}\pi R^3} \int_0^R r^4 dr \int_0^{\pi} \sin^3 \theta d\theta \int_0^{2\pi} d\phi$$

$$= \frac{M}{\frac{4}{3}\pi R^3} \left(\frac{R^5}{5} \right) \int_0^{\pi} (1-u^2) du (2\pi)$$

$$= \frac{2}{5} MR^2$$

3.33



"Square...
rotating about
perpendicular
axis"

(3.31) "Moment of Inertia"

$$I = \sum_{\alpha=1}^N m_{\alpha} r_{\alpha}^2$$

$$= \int_0^b r^2 dM$$

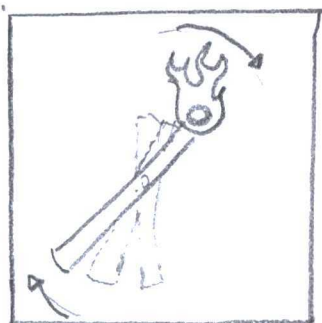
$$= \int_0^b r^2 \left(\frac{M}{A} dA \right)$$

$$= \frac{M}{4b^2} \int_0^b r^2 dA$$

$$= \frac{M}{4b^2} \int_0^b r^2 (4b dx)$$

$$= \frac{1}{3} Mr^2$$

3.34



"uniform rod... in tar and burning
... and its rotating with angular
velocity ω_0 "

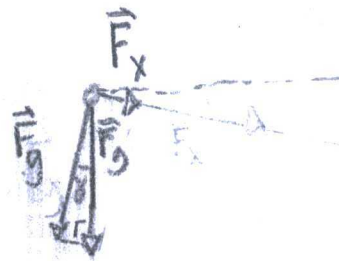
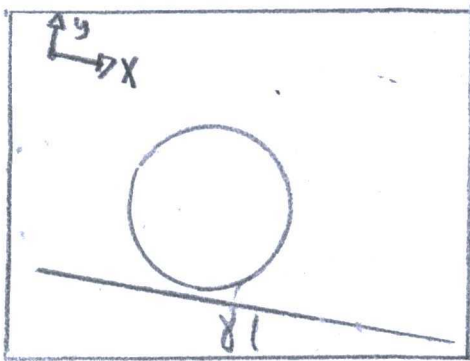
$$\omega_0 = \frac{v_0}{r}$$

$$n \omega_0 = \frac{v_0}{r} \cdot \frac{2\pi}{n}$$

$$v_0 = \frac{n \omega_0 r}{2\pi}$$

3.35

a) Free-body Diagram:



"Uniform solid disc...rolling without slipping down an incline"

b) Linear Acceleration ($\ddot{v}$):

(from problem) "Angular Momentum"

$$\dot{L} = \tau^{\text{ext}}$$

$$L = I\omega$$

$$= \frac{3}{2}MR^2$$

$$\text{So, } \dot{L} = I\dot{\omega}$$

$$= \frac{3}{2}MR^2\dot{\omega}$$

$$= \frac{3}{2}MR\dot{v}$$

$$\boxed{R\dot{\omega} = \dot{v}}$$

= "Force in x-direction"

$$= MgR \sin \theta$$

$$\dot{v} = \frac{2}{3} g \sin \theta$$

c) Note: With friction, the disc rotates unevenly. Chapter 10 explains moment of inertia in multiple frames.

$$\begin{aligned} I_{\text{TOTAL}} &= I_{\text{cm}} + I_{\text{rot}} \\ &= \frac{1}{2} MR^2 + MR^2 \end{aligned}$$

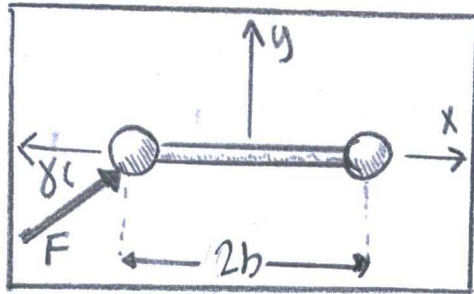
$$\begin{aligned} L &= I_{\text{TOT}} \omega \\ &= \left(\frac{1}{2} MR^2 + MR^2 \right) \omega \\ &= \frac{3}{2} MR \dot{v} \end{aligned}$$

= "Force in x-direction"

$$= MgR \sin \theta - F_{\text{friction}}$$

$$\dot{v} = \frac{2}{3} (1 - \mu) g \sin \theta$$

3.36



"Force acts in
a northeasterly
direction"

Velocity center of mass:

(From Example #4) "Impulse"

$$\vec{P} = M \dot{\vec{R}}$$

$$= 2m \dot{\vec{R}}$$

$$= \vec{F} \cdot \Delta t$$

$$\vec{V}_{cm} = \dot{\vec{R}}$$

$$= \frac{\vec{F} \Delta t}{2m}$$

Velocity at left dumbbell:

$$\vec{V}_{left} = \vec{V}_{cm} + \omega b$$

$$= \frac{\vec{F} \Delta t}{m}$$

$$= \frac{\sqrt{(F \sin \alpha)^2 + (F \cos \alpha)^2} \cdot \Delta t}{m}$$

$$= g \sqrt{(\sin \alpha)^2 + (\cos \alpha)^2} \Delta t$$

Velocity at right dumbbell:

$$\vec{V}_{right} = \vec{V}_{cm} - \omega b$$

$$= 0$$

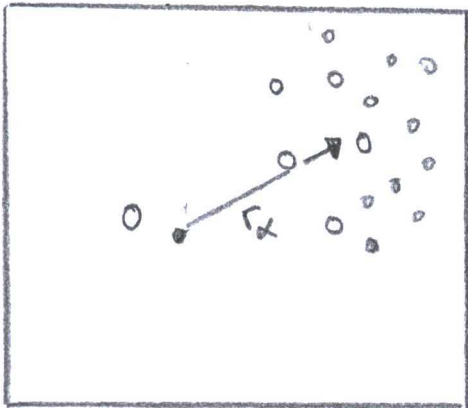
If $\gamma = 0^\circ$, then $v_{\text{left}} = g \cdot \Delta t$

$$v_{\text{right}} = 0$$

If $\gamma = 90^\circ$, then $v_{\text{left}} = g \cdot \Delta t$

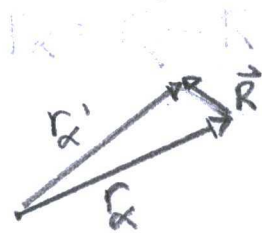
$$v_{\text{right}} = 0$$

3.37.



"A system consists of N masses at positions r_α "

a) (from problem) "Relative distance"



$$\vec{r}'_\alpha = \vec{r}_\alpha - \vec{R}$$

b) Relative center:

$$\sum m_\alpha r'_\alpha = \sum m_\alpha (r_\alpha - R)$$

$$= \sum m_\alpha r_\alpha - \sum m_\alpha R$$

$$= M r_\alpha - M R$$

$$= 0 \quad @ \quad r'_\alpha \text{ is small relative to center of mass}$$

c) (3.28) "Angular Momentum"

$$\frac{d}{dt} L = \tau^{\text{ext}}$$

$$= \frac{d}{dt} (I \omega)$$

$$= \frac{d}{dt} \left(m R^2 \left[\frac{v}{R} \right] \right)$$

$$= \frac{d}{dt} (m R v)$$

$$= R m \frac{d}{dt} v$$

$$= R \cdot m a$$

$$= R \cdot F \quad \dots \text{torque}$$